Neuro-Forget

Explainable Alzheimer's Detection via Machine Unlearning on Blood Transcriptomics

Course: Bioinformatics Programming and Scripting | DS307

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1. Project Overview & Core Idea

Neuro-Forget is a privacy-preserving bioinformatics pipeline designed to detect Alzheimer's disease from blood gene expression data.
Unlike traditional genomic pipelines that operate on raw DNA sequences, this system leverages transcriptomics (gene expression) to build an interpretable and scalable AI model.
The system integrates Machine Unlearning (SISA) to support GDPR-compliant removal of patient data without retraining the entire model.

Key Objectives:

- Process transcriptomic gene expression data from GEO datasets
- Train machine learning models to classify Alzheimer's disease
- Explain model predictions using SHAP (interpretable AI)

- Implement SISA Machine Unlearning for efficient patient deletion
- Evaluate privacy risks using Membership Inference Attacks (MIA)

2. Team Distribution — Roles & Tasks

Focus Area A: Data Engineering & Preprocessing

Task	Description
Data Parsing	Load GEO dataset (GSE63060) and extract gene expression matrix
Label Extraction	Identify Alzheimer's vs Control samples from metadata
Feature Selection	Reduce dimensionality to most informative genes
Normalization	Standardize gene expression values
Dataset Structuring	Generate X (features) and y (labels)

Focus Area B: Modeling, Unlearning & Evaluation

Task	Description
Model Development	Train XGBoost and LightGBM classifiers
Unlearning Implementation	Apply SISA for efficient deletion
Privacy Verification	Use Membership Inference Attack
Performance Metrics	Evaluate Accuracy, F1, ROC-AUC, CV

3. Dataset Description — Alzheimer's Transcriptomics

3.1 Dataset Source

- Dataset: GSE63060 (GEO)
- Type: Blood gene expression (microarray)
- Disease: Alzheimer's disease

3.2 Dataset Statistics

Parameter	Value
Samples	329
Features	136 genes
Classes	AD vs Control

3.3 Labeling Strategy

Alzheimer's samples identified from metadata.
Binary encoding:

- 1 → Alzheimer's
- 0 → Control

3.4 Output Files

File	Description
X.csv	Gene expression features
y.csv	Labels
metadata	Sample annotations

4. System Workflow

Step	Stage	Description
1	Input	GEO transcriptomics dataset
2	Preprocessing	Cleaning, normalization, feature selection
3	Model Training	XGBoost / LightGBM training
4	Prediction	Disease classification
5	Explainability	SHAP feature importance
6	Deletion Trigger	Patient removal request
7	Unlearning	Retrain only affected shard
8	Verification	Privacy audit (MIA)

5. Model Architecture & Training

Models Used

- XGBoost
- LightGBM

Why these models?

- Strong performance on tabular biological data
- Handle non-linear relationships
- Robust to noise

Training Strategy

- Train-test split (stratified)
- 5-fold cross-validation
- Hyperparameter tuning

6. Results & Performance

Model	Accuracy	F1 Score	ROC-AUC	CV ROC-AUC
XGBoost	0.712	0.667	0.797	0.785 ± 0.058
LightGBM	0.712	0.678	0.804	0.768 ± 0.064

Key Insights

- ROC-AUC ≈ 0.80 → strong classification ability
- LightGBM slightly outperforms XGBoost
- Cross-validation confirms model stability
- No major overfitting observed

7. Explainability (SHAP Analysis)

SHAP (SHapley Additive Explanations) was used to interpret model predictions.
Top important genes:

- ILMN_1776104
- ILMN_1784286
- ILMN_1749834
- ILMN_2066060

Insight

- Model is interpretable, not black-box
- Provides biological relevance
- Useful for biomarker discovery

8. Machine Unlearning (SISA)

Approach

- Data divided into shards
- Each shard trained independently
- Aggregated into final model

Deletion Process

- Identify patient shard
- Remove patient data
- Retrain only that shard
- Update global model

Benefits

- Efficient
- Scalable
- GDPR-compliant

9. Privacy Validation

Method

- Membership Inference Attack (MIA)

Findings

- Some confidence leakage observed
- Indicates potential memorization
- Highlights importance of privacy-aware AI

10. System Integration & Deployment

Streamlit UI Features

- Model selection (XGBoost / LightGBM)
- Dataset visualization
- Prediction interface
- SHAP explainability
- Patient deletion
- Privacy audit

11. Grading Rubric Alignment

Criteria	Description
Technical Complexity	Real biomedical data pipeline
AI Model Performance	ROC-AUC ≈ 0.80
Unlearning Logic	Efficient SISA implementation
System Integration	Full pipeline + UI
Privacy Validation	MIA implemented
Documentation	Clear report + code

12. Limitations

- Moderate dataset size
- Blood data ≠ brain tissue
- No deep learning models
- Limited external validation

13. Future Work

- Multi-dataset integration
- Deep learning models (TabNet, Transformers)
- Single-cell RNA-seq
- Clinical deployment

FINAL NOTE

This project demonstrates a complete AI system in bioinformatics, combining:

- Machine Learning
- Explainable AI
- Privacy Preservation
- Real-world deployment